

Review

Low carbon concrete: advancements, challenges and future directions in sustainable construction

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Abstract

Concrete production accounts for approximately 8% of global carbon emissions, underscoring the critical need for innovative low-carbon alternatives to support sustainable construction. This review addresses the environmental impact of conventional concrete and its substantial carbon footprint, emphasizing the urgency for transformative technologies. The study examines emerging low-carbon concrete solutions, including the incorporation of supplementary cementitious materials (SCMs), alkali-activated and geopolymer concrete, carbon capture and utilization (CCU), and recycled aggregates, which collectively show potential to reduce emissions by up to 30–50%. The paper also explores advanced mix design strategies aimed at minimizing emissions while maintaining or enhancing structural performance, which could improve durability by 20–25% and decrease lifecycle costs by up to 15%. Challenges to large-scale implementation are critically analyzed, including regulatory obstacles, technical constraints, and market dynamics, with a particular focus on achieving widespread market acceptance within a decade. Future research directions, such as material innovations, enhanced carbon capture methods, and the integration of digital technologies like AI for predictive modelling, are highlighted as pivotal for global adoption. This review demonstrates that the adoption of low-carbon concrete technologies is essential to achieving international climate targets, such as limiting global warming to 1.5 °C while contributing to the construction sector's sustainability. The findings provide a robust foundation for international research collaboration and policy development aimed at decarbonizing the construction industry.

Keywords Low-carbon concrete · Sustainable construction · Carbon footprint reduction · Supplementary cementitious materials (SCMs) · Carbon capture and utilization (CCU)

1 Introduction

Climate change, driven by greenhouse gas emissions, is increasingly impacting the construction sector. The industry contributes significantly to emissions through energy use, material production, and waste generation. Buildings and infrastructure account for a large portion of global carbon footprints [1–4]. As climate change accelerates, construction practices face growing pressures to reduce environmental impact and adapt to changing conditions. This includes adopting energy-efficient technologies, sustainable materials, and resilient designs to withstand extreme weather events

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[5–9]. The sector is critical in the fight against climate change, with the potential for substantial emissions reductions and enhanced sustainability through innovative practices and regulatory changes.

Concrete is a major contributor to global carbon emissions, accounting for about 8–10% of total emissions [10–12]. Its production process involves cement, a key ingredient that emits significant CO₂ when limestone is heated to produce clinker. The widespread use of concrete in construction, from buildings to infrastructure, amplifies its environmental impact. Efforts to mitigate this include developing low-carbon cement, incorporating recycled materials, and optimizing mix designs [13–16]. Despite advances, the sector faces challenges in reducing emissions due to concrete's essential role in modern construction. Innovations and stricter regulations are crucial to addressing the environmental footprint of concrete.

The need for low-carbon concrete technologies is urgent due to the significant carbon footprint of traditional concrete production. Conventional concrete emits substantial CO₂ through cement manufacturing, a process that heats limestone to produce clinker. To combat climate change, it's crucial to develop and adopt low-carbon concrete alternatives. Innovations include using alternative materials like fly ash or slag, enhancing mix designs, and incorporating carbon capture technologies [17–20]. These technologies can reduce emissions, improve sustainability, and support the construction industry's transition to greener practices. Embracing low-carbon concrete is essential for mitigating environmental impact and achieving global climate goals.

Low-carbon concrete has emerged as a vital innovation to reduce the carbon footprint of construction activities. Recent research highlights the potential of incorporating waste materials into concrete formulations to achieve environmental and mechanical performance goals. Tahwia et al. [21] demonstrate that high-strength concrete reinforced with alkali-resistant (AR) glass fibers and incorporating high volumes of solid waste materials exhibits significant durability under environmental stressors. This approach reduces reliance on energy-intensive cement production while improving long-term performance. Geopolymer composites, another avenue of sustainable concrete, show promise for their physical and mechanical robustness. Tahwia et al. [22] explored eco-friendly engineered geopolymer composites, emphasizing their enhanced durability and reduced carbon emissions. These materials offer a sustainable alternative to traditional cement by utilizing industrial by-products such as fly ash and slag. In another study, Abdellatif et al. [23] investigated alkali-activated concrete incorporating waste medical glass, finding that such mixtures perform well under carbonation and elevated temperatures. This underscores their suitability for diverse environments while addressing waste management challenges. Additionally, advanced predictive modelling of foam concrete compressive strength, as analyzed by Abdellatif et al. [24], highlights the role of artificial intelligence in optimizing material properties and design, paving the way for innovative low-carbon construction solutions.

The objective of this review is to evaluate and synthesize advancements in technologies and practices aimed at minimizing the carbon footprint of concrete production. It assesses low-carbon alternatives, such as innovative materials, mix designs, and production techniques, focusing on their effectiveness in reducing CO₂ emissions. The review covers recent innovations, comparative analyses, and their implications for the construction industry, emphasizing current research, challenges, and opportunities. By providing a holistic perspective, this study aims to enhance understanding of low-carbon concrete's role in promoting sustainable construction and supporting climate mitigation objectives.

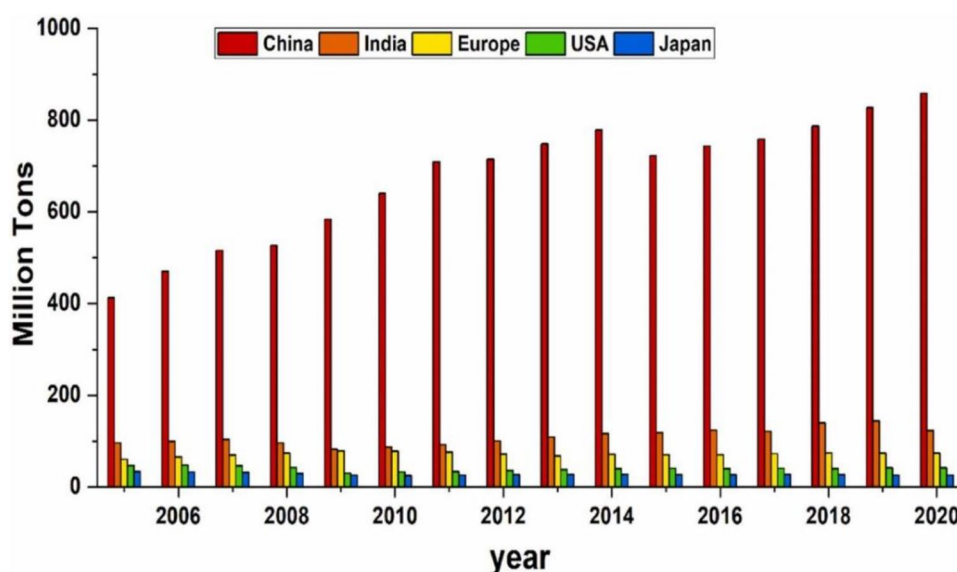
2 Conventional concrete and its carbon footprint

Traditional concrete production involves mixing cement, water, and aggregates (sand, gravel, or crushed stone) to create a durable building material. Cement, the primary component, is produced by heating limestone and other raw materials in a kiln to create clinker, which is then ground into a fine powder. This process is highly energy-intensive and contributes significantly to CO₂ emissions, as both limestone decomposition and fuel combustion release greenhouse gases [25, 26]. The widespread use of concrete in construction is due to its strength and versatility, but its environmental impact highlights the urgent need for more sustainable production methods.

The raw materials for traditional concrete—cement, sand, gravel, and water—each have notable environmental impacts. Cement production is particularly carbon-intensive, with the heating of limestone to produce clinker emitting substantial amounts of CO₂. Sand and gravel extraction can lead to habitat destruction, water pollution, and soil erosion, while the energy required for mining and transporting these materials further contributes to emissions. Addressing these issues requires adopting more sustainable practices, such as using alternative materials or recycled content and improving resource management. Although efforts are underway to develop alternative cement, enhance energy efficiency, and incorporate carbon capture technologies, significant challenges remain in reducing the overall environmental impact

Table 1 Global and regional overview of cement production and CO₂ emissions

Region/country	Annual cement production (metric tons)	CO ₂ emissions from cement production (metric tons)	CO ₂ Emissions per ton of cement (kg CO ₂ /ton)	References
Global average	4.1 billion	2.8 billion	680	[27]
USA	90 million	60 million	667	[28]
China	2.4 billion	1.8 billion	750	[29]
India	330 million	250 million	758	[30]
Europe (EU-28)	200 million	130 million	650	[31]

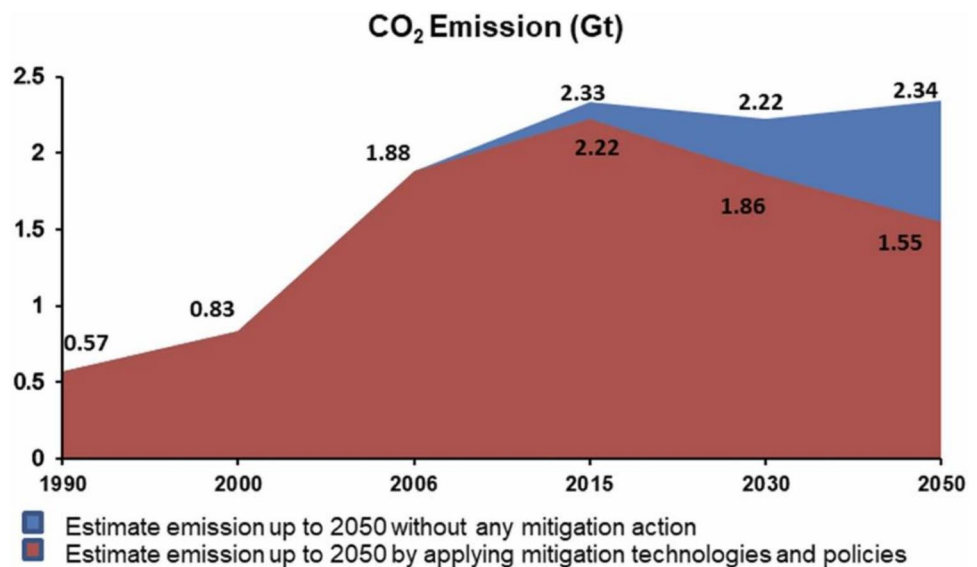
Fig. 1 Top countries with the highest CO₂ emission annually from the cement industry (2005–2019) [35]

of concrete production. Table 1 provides an overview of cement production and associated CO₂ emissions for different regions or countries, including the global average. The table gives a snapshot of production volume, total emissions, and emissions intensity per ton of cement, which can be useful for understanding regional variations in the environmental impact of cement production.

Quantifying carbon emissions from conventional concrete involves assessing the entire lifecycle of concrete production, including raw material extraction, transportation, mixing, and curing. Cement production, a key component, is responsible for a significant portion of these emissions due to the high energy required and the chemical reactions involved. Typically, each ton of cement emits around 680–750 kg of CO₂ [32–34]. Factors such as the efficiency of production processes, the use of alternative fuels, and the incorporation of supplementary materials can influence the overall carbon footprint. Accurate quantification is crucial for developing strategies to mitigate emissions and enhance sustainability in the construction industry.

In 2020, global CO₂ emissions were significantly contributed by China and India, accounting for an estimated 35%, while the UK, Germany, and France together contributed only 4% [26]. Between 2005 and 2021, the highest CO₂ emitters were China, India, Europe, and the USA (Fig. 1) [35]. During the COVID-19 pandemic, emissions from the cement sector remained relatively stable, showing less decline compared to sectors like oil, gas, and coal [36]. However, by implementing strategies such as using alternative fuels, improving energy efficiency, substituting clinker, and employing carbon capture and utilization (CCU), cement industry CO₂ emissions could be reduced to 1.5 billion tons per year and 0.43 tons of CO₂ per ton of cement by 2050 (Fig. 2) [37]. Heidelberg Cement has committed to reducing CO₂ emissions by 25–30% by 2030 compared to 1999 levels and was the first company to receive approval for science-based CO₂ capture targets. Various CO₂ capture technologies, including amine scrubbing, calcium looping, oxyfuel combustion, and direct separation, have been evaluated. However, pre-combustion processes are not ideal for the cement industry, as they are better suited for capturing energy-related CO₂ emissions [38].

Fig. 2 Globally CO₂ emission from 1990 to 2050 with and without any mitigation strategies [37]



3 Low carbon concrete: definitions and concepts

Low-carbon concrete is a sustainable alternative to traditional concrete that significantly reduces carbon dioxide emissions during its production. Unlike conventional concrete, which relies heavily on Portland cement—a major source of CO₂ emissions—low-carbon concrete uses materials like fly ash, slag, or alternative cement to lower its carbon footprint. These materials not only reduce greenhouse gas emissions but also often enhance the concrete's performance and durability. By incorporating waste products or less carbon-intensive ingredients, low-carbon concrete contributes to greener construction practices and helps mitigate climate change, aligning with global efforts to create more sustainable infrastructure.

Carbon footprint reduction strategies aim to minimize greenhouse gas emissions across various sectors. Key strategies include adopting renewable energy sources like solar or wind, improving energy efficiency in buildings and industrial processes, and transitioning to low-carbon transportation options such as electric vehicles [39–42]. Additionally, practices such as reducing waste, recycling materials, and using energy-efficient appliances contribute to lowering emissions. Supporting sustainable agriculture and conserving natural habitats also play crucial roles. By integrating these strategies, individuals and organizations can significantly reduce their carbon footprints, helping to combat climate change and promote a more sustainable future.

Performance criteria for low-carbon concrete alternatives focus on achieving both environmental sustainability and practical functionality in construction [43–45]. To meet structural requirements, compressive strength is a primary consideration; low-carbon concrete must attain strength levels that match or exceed traditional concrete, ensuring it can withstand load-bearing demands safely. Durability is also critical, as the material must resist environmental factors such as temperature fluctuations, moisture, and potential chemical exposure to avoid premature deterioration. Workability is another essential factor, as it affects ease of placement and finishing during construction. A low-carbon alternative should exhibit adequate flow and consistency to ensure it can be handled similarly to conventional concrete, minimizing challenges for construction teams. The environmental impact, primarily measured through lifecycle analysis, is central to low-carbon criteria. These materials should offer a significantly reduced carbon footprint compared to traditional options, aiding efforts to reduce overall emissions in the construction sector. Moreover, cost-effectiveness and compatibility with existing construction techniques are important for broader adoption. If low-carbon alternatives align with current practices and are economically viable, they become more practical for industry-wide implementation, supporting sustainability goals without compromising on performance or affordability.

4 Technological innovations in low carbon concrete

4.1 Supplementary cementitious materials (SCMs)

Fly ash, slag, and silica fume are supplementary cementitious materials (SCMs) that enhance the properties of concrete and help reduce its environmental impact. Fly ash, a byproduct of coal combustion, improves concrete's workability and strength over time. Slag, which comes from blast furnace waste, enhances durability and lowers permeability. Silica fume, a byproduct of silicon metal production, increases concrete's density and strength due to its fine particles. By using these SCMs, the amount of Portland cement needed in concrete is reduced, which in turn lowers CO₂ emissions from cement production. This not only makes construction practices more sustainable but also boosts the performance and longevity of concrete structures.

The use of SCMs in low-carbon concrete is a promising approach to reducing the environmental impact of cement production. SCMs such as fly ash, silica fume, ground granulated blast-furnace slag (GGBS), and recycled quarry dust provide alternative binder materials that can partially replace Portland cement, thereby lowering carbon emissions [46, 47]. Their incorporation not only reduces the clinker content—significantly lowering CO₂ emissions—but also often enhances concrete's durability and strength at later ages [48]. However, there are challenges. While materials like fly ash and GGBS improve long-term strength, they can compromise early-age strength, potentially limiting their application in projects requiring quick construction turnaround [49]. Advances in nano-modification and interfacial engineering, such as the use of nano-TiO₂ with fly ash, have shown promise in enhancing early strength, but these methods can increase costs and complexity [50]. Additionally, recycling construction waste for SCMs, such as using CO₂-mineralized demolished concrete, presents a dual benefit of waste reduction and carbon sequestration, yet requires substantial processing [51]. Cement-free binders, although innovative, face scalability issues and require further development to ensure they meet performance standards consistently [52]. Thus, while SCMs hold potential, balancing performance, cost, and environmental benefits remains a critical challenge.

SCMs are essential for minimizing the carbon footprint of concrete. They allow for a reduction in the overall cement content, which leads to decreased CO₂ emissions. Additionally, SCMs often repurpose industrial byproducts, helping to divert waste from landfills and reduce resource extraction. While SCMs improve concrete properties like strength and durability, their effectiveness can vary, potentially affecting concrete performance and curing times. Moreover, integrating SCMs into concrete mixes can have economic implications. A careful mix of design and quality control is crucial to ensure SCMs' benefits are fully realized while meeting regulatory standards and maintaining consistent performance.

Johari et al. [53] found that silica fume and metakaolin improved concrete strength, while fly ash and GGBS enhanced long-term strength despite reducing early-age strength as can be seen in Fig. 3. The figure illustrates the relative compressive strength development over time for different low-carbon cementitious materials compared to ordinary Portland cement (OPC). Each subplot shows a different material: (a) silica fume (SF), (b) fly ash (FA), (c) metakaolin (MK), and (d) ground granulated blast-furnace slag (GGBS). For silica fume (a), compressive strength surpasses OPC quickly, especially at higher dosages (SF10 and SF15), with strength slightly decreasing over time but remaining above OPC. Fly ash (b) shows a slower strength gain initially but reaches or surpasses OPC's strength at later ages (after 28 days), with higher FA content resulting in higher long-term strength. Metakaolin (c) achieves early strength gains, especially at higher dosages (MK10 and MK15), consistently exceeding OPC's strength throughout the test period. GGBS (d) shows a slower early-age strength gain but achieves comparable or higher long-term strength, especially at higher replacements (GGBS40 and GGBS50). Overall, materials like SF and MK enhance early strength, while FA and GGBS contribute to long-term strength improvements, demonstrating diverse benefits for different construction needs.

Natural pozzolans, ceramic slag, and rice-husk ash are sustainable alternatives to traditional cementitious materials in concrete, promoting low-carbon construction practices. These materials contribute to reducing the carbon footprint of concrete by partially replacing Portland cement, a significant source of CO₂ emissions during its production. Natural pozzolans, such as volcanic ash and pumice, are rich in silica and alumina, which react with calcium hydroxide in the presence of water to form additional cementitious compounds [54, 55]. This enhances the strength and durability of concrete while minimizing reliance on energy-intensive clinker production. Ceramic slag, a byproduct of ceramic and tile manufacturing, contains high levels of silica, alumina, and other oxides. When ground into fine particles, it serves as a SCM, improving the concrete's mechanical properties and durability [56, 57]. Utilizing ceramic

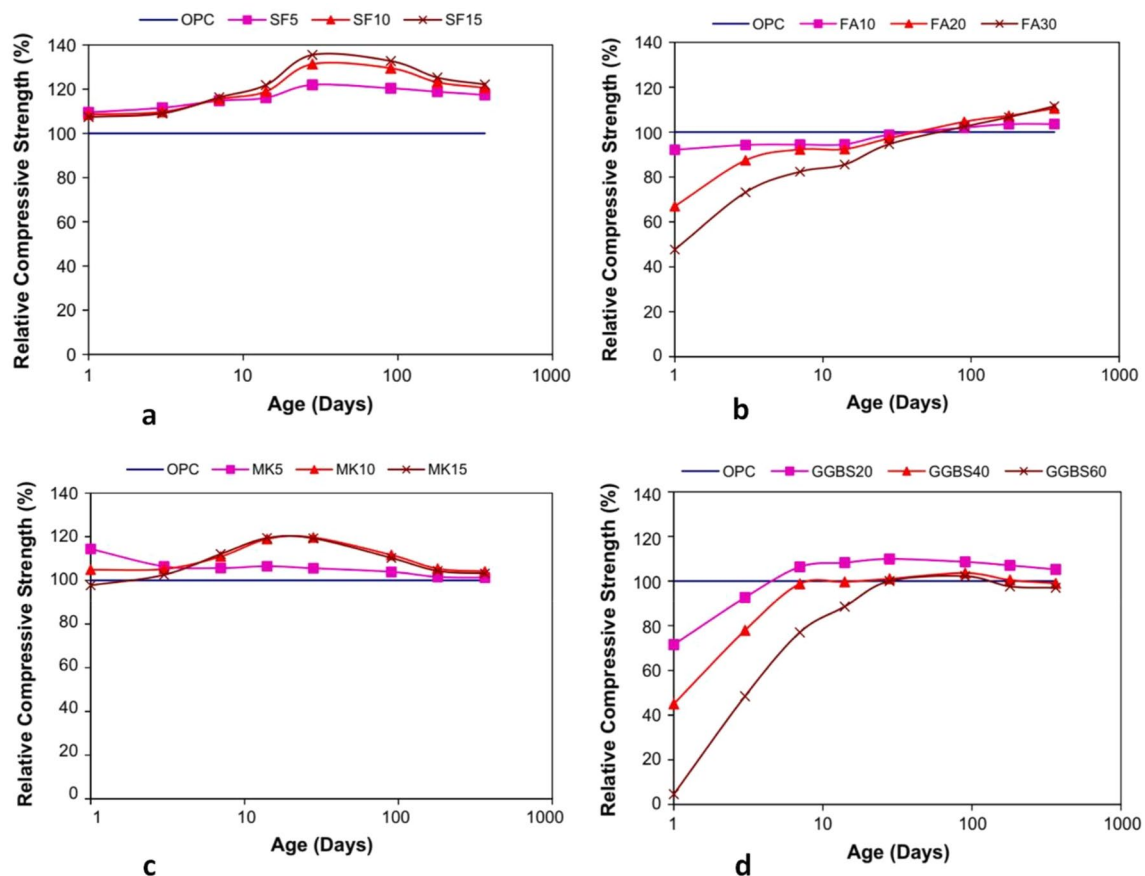


Fig. 3 Effect of different SCMs on relative concrete compressive strength up to 365 days [53] (a) silica fume; (b) Fly ash; (c) Metakaolin; and (d) GGBS

slag reduces waste and diverts it from landfills, aligning with circular economy principles. Rice-husk ash, derived from burning rice husks, is a rich source of amorphous silica. When processed and incorporated into concrete, it enhances compressive strength, reduces permeability, and offers significant resistance to chemical attacks [58, 59]. Its utilization supports agricultural waste valorization and reduces dependence on natural resources. These materials exemplify eco-friendly innovations in concrete, promoting sustainable construction and addressing global carbon reduction goals.

4.2 Alkali-activated and geopolymer concrete

Alkali-activated and Geopolymer concrete are cutting-edge materials produced by activating aluminosilicate precursors with alkaline solutions. In alkali-activated concrete, materials like fly ash or slag are combined with strong alkaline solutions such as sodium hydroxide or potassium hydroxide to create a binder [60, 61]. Geopolymer concrete follows a similar process but often utilizes metakaolin or other aluminosilicate sources. The activation forms a three-dimensional network of aluminosilicate gel, which imparts high strength and durability to the concrete [62–64]. These innovative materials offer an eco-friendly alternative to traditional Portland cement, featuring a reduced carbon footprint and enhanced performance characteristics.

Alkali-activated and geopolymer concretes represent a significant low-carbon alternative to conventional cement, reducing reliance on clinker by up to 80% and enabling the utilization of industrial byproducts such as fly ash, slag, and red mud [65, 66]. These materials derive their binding properties from a chemical reaction (geopolymerization) between aluminosilicate precursors and alkali activators, achieving compressive strengths often exceeding 40–60 MPa, with excellent durability and resistance to environmental degradation [67, 68]. Studies indicate that geopolymer concrete exhibits superior acid resistance and thermal stability, making it particularly suitable for use in corrosive and high-temperature environments [69].

However, alkali-activated and geopolymer concretes face challenges related to consistency and performance variability, with sensitivity to precursor quality, mix design, and curing conditions influencing outcomes by as much as 30–50% [62]. The choice of alkali activators, such as sodium silicate or sodium hydroxide, significantly impacts the geopolymerization process, with variations contributing to differences in setting time, workability, and strength development [70, 71]. For instance, fly ash sourced from various industrial processes can yield inconsistent results, with compressive strength varying by 20–40% based on precursor composition and reaction conditions.

The environmental benefits of geopolymer concrete are contingent on the production and sourcing of alkali activators, which can contribute up to 10–20% of the total carbon footprint of the material. Advanced mix design strategies, incorporating artificial intelligence, aim to optimize compressive strength (targeting ranges of 50–70 MPa), cost, and carbon emissions. Nevertheless, scalability remains a major hurdle, with production costs and logistical challenges increasing by 15–25% compared to traditional cement [72]. Furthermore, the high pH of alkali activators presents safety concerns, particularly in large-scale applications, requiring stringent handling protocols [73]. Enhancing the mechanical properties of geopolymer concretes through additives such as nano-silica or steel fibres has demonstrated up to a 20–30% improvement in performance under dynamic and explosive loading conditions. However, these modifications can elevate costs by approximately 10–15% [74, 75]. To achieve widespread adoption, it is essential to address these economic and technical challenges, focusing on standardization, sustainability, and the development of consistent and scalable production methods.

Zhang et al. [76] reviewed the fresh properties and durability of Geopolymer Concrete (GPC). Figure 4 comprises two graphs. The left bar chart illustrates the weight loss percentage of different concrete mixes—Ordinary Portland Cement (OPC) and five Geopolymer Concrete (GPC) variants—over time intervals of 56, 90, 180, 270, and 365 days. OPC shows significantly higher weight loss, indicating poor durability compared to GPC mixes, which exhibit lower weight loss percentages, suggesting better chemical resistance. The right graph is a line chart showing the residual compressive strength (MPa) of the same mixes over the same time intervals. OPC exhibits a sharp decrease in strength over time, especially beyond 90 days. In contrast, all GPC variants maintain relatively stable residual compressive strengths with minor reductions, demonstrating superior durability and long-term performance under adverse conditions. Overall, the data suggest that GPC outperforms OPC in terms of weight loss resistance and residual compressive strength, making it a more durable and resilient material in the long term.

Figure 5 contains two graphs comparing Geopolymer Concrete (GPC) with different silica fume (SF) contents (0%, 10%, 20%) to Ordinary Portland Cement (OPC) based on weight loss and compressive strength loss over exposure periods of 3, 28, 56, and 90 days. The left bar chart illustrates weight loss (%). OPC exhibits higher weight loss, increasing with exposure time, indicating lower resistance to degradation. GPC mixes show significantly lower weight loss, with improved performance as SF content increases, highlighting the positive impact of SF on durability. The right line graph depicts compressive strength loss (%). OPC demonstrates a steep increase in strength loss over time, reflecting poor durability. GPC with 10% and 20% SF exhibits negligible strength loss, maintaining stability over the

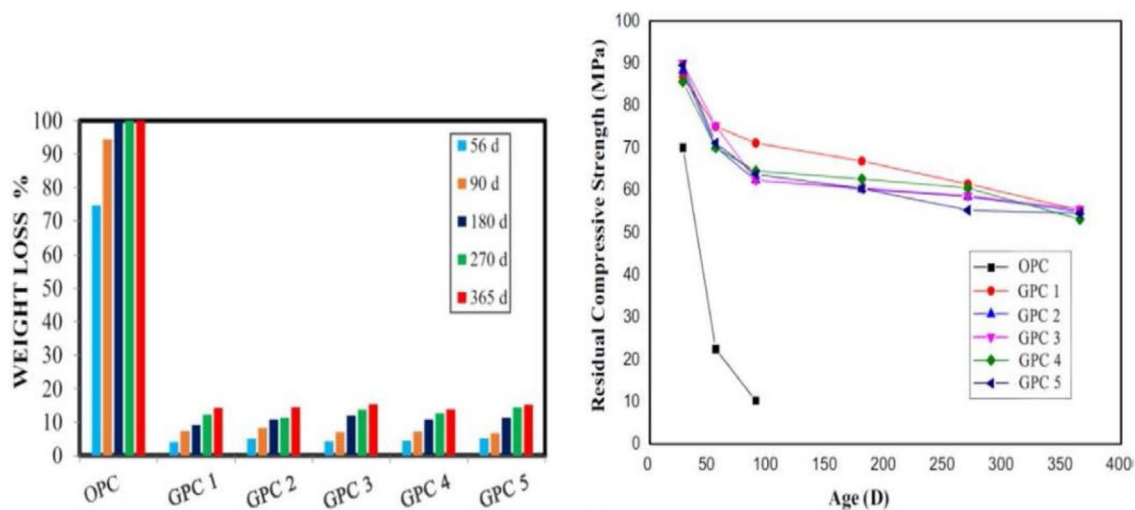


Fig. 4 Weight loss and compressive strength of GPC and OPC exposed to 5% H_2SO_4 for different durations [76]

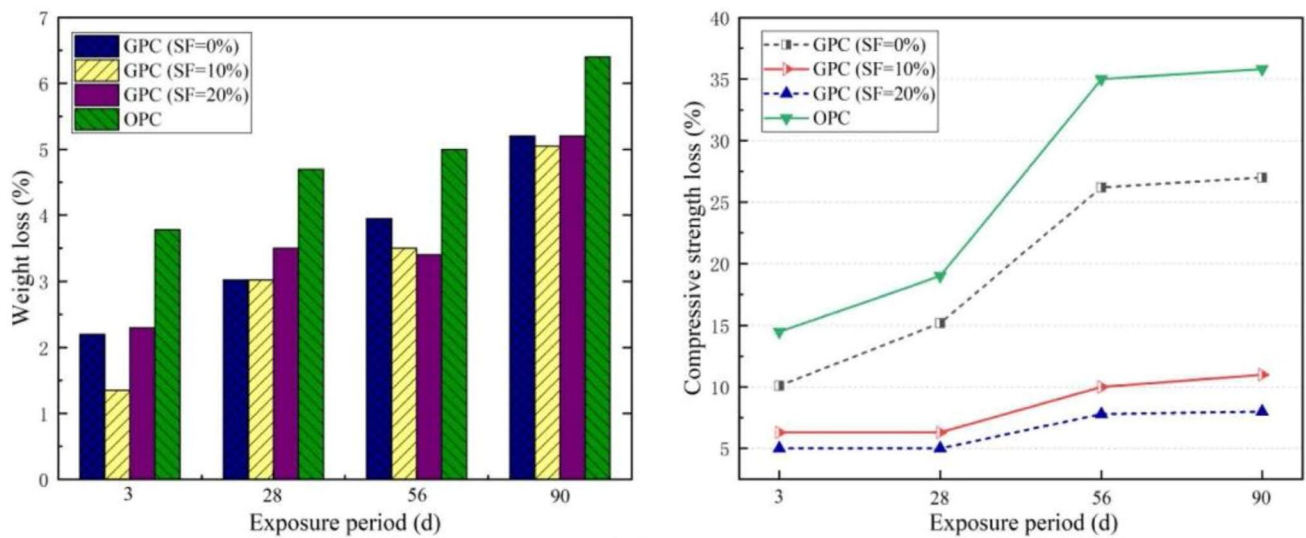


Fig. 5 Weight loss and compressive-strength loss of GPC and OPCC exposed to 2% H_2SO_4 for different durations [76]

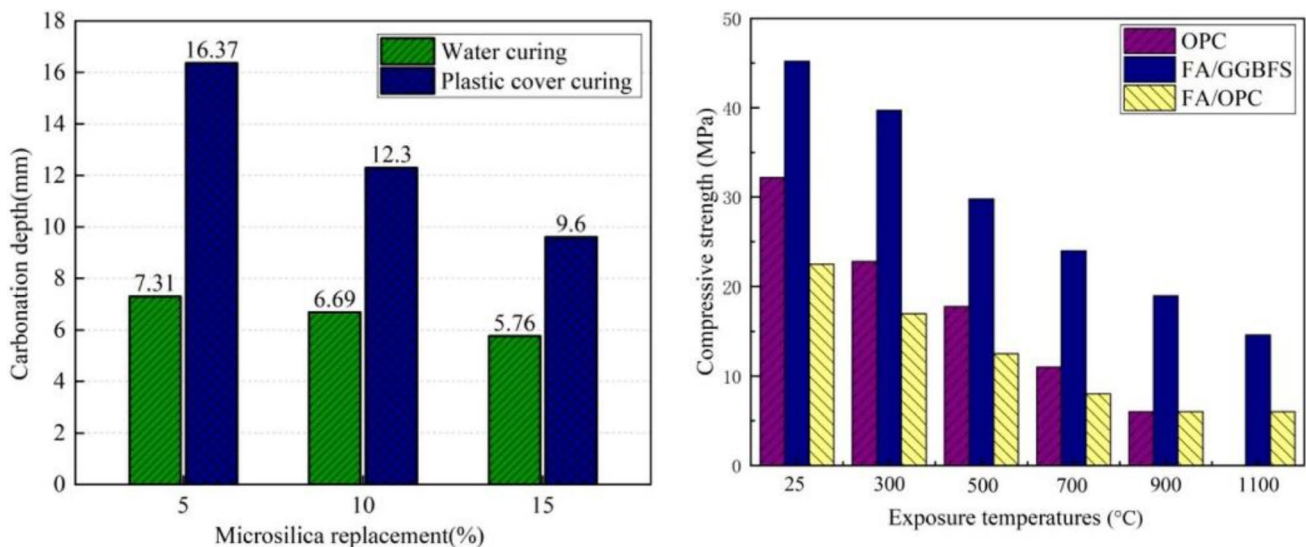


Fig. 6 Carbonation and temperature resistance of GPC [76]

exposure period. GPC without SF shows slightly higher strength loss but still outperforms OPC. Overall, GPC with SF provides superior durability, resisting both weight loss and compressive strength degradation better than OPC.

Figure 6 consists of two bar charts. The first chart compares carbonation depth (in mm) under two curing methods—water curing and plastic cover curing—at different levels of microsilica replacement (5%, 10%, and 15%). Water curing consistently results in lower carbonation depth compared to plastic cover curing, indicating better performance in carbonation resistance. The carbonation depth decreases with higher microsilica replacement for both curing methods, showcasing the positive impact of microsilica in reducing carbonation susceptibility. The second chart examines compressive strength (in MPa) for different materials (OPC, FA/GGBFS, and FA/OPC) exposed to varying temperatures (25 °C to 1100 °C). OPC maintains the highest compressive strength across all temperatures but declines significantly at elevated temperatures. FA/GGBFS exhibits moderate strength and better thermal resistance, while FA/OPC has the lowest strength overall. The results highlight OPC's initial strength superiority and the better high-temperature stability of FA/GGBFS blends.

4.3 Carbon capture and utilization (CCU) in concrete

Carbon sequestration in concrete involves capturing and storing carbon dioxide (CO₂) within the material, mitigating its greenhouse gas emissions. This process typically occurs through mineral carbonation, where CO₂ reacts with calcium silicates in the concrete, forming stable calcium carbonate. This not only reduces atmospheric CO₂ but also improves concrete durability and strength. Techniques include using CO₂ in the curing process or incorporating industrial by-products like fly ash or slag, which enhance CO₂ absorption. Innovations in this area promise to transform concrete production into a more sustainable practice, potentially offsetting a significant portion of the construction industry's carbon footprint.

Carbon mineralization converts CO₂ into stable solid minerals through chemical reactions. In situ mineralization involves injecting CO₂ into underground rock formations rich in calcium, magnesium, or iron, where it reacts to form stable carbonates. Ex-situ mineralization captures CO₂ and reacts it with mineral ores or industrial by-products in a controlled environment, also producing carbonates. Bio-mineralization utilizes microorganisms to facilitate the reaction between CO₂ and minerals, enhancing carbonate formation. Each method has its benefits, such as scalability and cost-effectiveness, and contributes to climate change mitigation by providing a long-term solution for CO₂ storage.

The integration of Carbon Capture and Utilization (CCU) into concrete production represents an innovative approach to reducing the cement industry's significant carbon footprint, yet challenges persist in achieving net climate benefits and economic feasibility. CCU in concrete primarily involves the mineralization of CO₂, wherein CO₂ is chemically reacted with calcium ions from cementitious materials, either during concrete curing or by mineralizing recycled aggregates [77]. Studies indicated that such processes can enhance the durability and strength of concrete, providing performance benefits that may reduce the total material demand [78]. However, the overall climate impact remains complex to assess due to variable energy and material inputs.

While CCU holds promise for reducing emissions in theory, the actual environmental impact varies with specific CCU applications. Ravikumar et al. [79] showed that CCU in concrete can mitigate some emissions but does not always result in a net benefit when the entire life cycle is considered, particularly due to energy-intensive CO₂ capture processes. Furthermore, Gálvez-Martos et al. [80] found that the eco-efficiency of CCU-based materials is highly dependent on the purity and source of CO₂, as well as the logistics involved in CO₂ transport. A holistic life cycle analysis, as conducted by Ang et al. [81], demonstrated that while the CCU process does sequester CO₂, it can also introduce indirect emissions, particularly in the CO₂ capture, transport, and injection phases.

Feasibility remains a critical barrier, with studies like Monteiro and Roussanly [82] questioning the economic viability of CCU in cement due to high capital and operational costs. The carbon price and the availability of CO₂ sources are crucial factors affecting deployment, as illustrated in region-specific studies like those of Abdelshafy and Walther [83] for Western Germany. Additionally, long-term carbon storage in concrete raises concerns, as some CO₂ can be re-released over time, diminishing the net climate benefit [84, 85]. While CCU in concrete has the potential to reduce emissions, realizing substantial climate benefits requires further optimization of capture methods, utilization efficiency, and supply chains [86, 87]. Integrative policies and incentives are essential to offset costs, as CCU alone may be insufficient to achieve the industry's ambitious carbon neutrality goals without complementary low-carbon strategies in concrete production.

Table 2 summarizes the methods, their descriptions, advantages, and challenges, providing a clear overview of each approach to carbon mineralization. These methods can reduce the carbon footprint of construction projects and improve concrete performance. However, limitations include high costs, the need for infrastructure to capture and transport CO₂, and the limited availability of suitable industrial by-products. Additionally, the scalability of CCU technologies remains a challenge, impacting their widespread adoption in the construction industry.

The pathways of mineral carbonation are shown in Fig. 7 [88]. The figure provides an overview of mineral carbonation methods for carbon capture and utilization, divided into direct and indirect carbonation processes. In direct carbonation, two main approaches are highlighted: gas–solid carbonation, which involves exposing minerals like olivine, wollastonite, and serpentine to CO₂ at high pressures (100–150 bar) to form stable carbonates, and aqueous carbonation, where CO₂ is dissolved in a solution that reacts with minerals under controlled pressure and temperature. The result is the permanent fixation of CO₂ as carbonates. Indirect carbonation involves extracting calcium and magnesium from waste materials using various acids (HCl, HNO₃, CH₃COOH, or NH₄Cl). The extracted Ca/Mg solution reacts with CO₂ to produce pure calcium carbonate, which has applications in construction. This process not only sequesters CO₂ but also generates useful byproducts, such as metals and silicates. Both carbonation pathways present sustainable solutions for utilizing CO₂ emissions, turning them into valuable construction materials while reducing greenhouse gases.

Table 2 Overview of carbon mineralization methods: benefits and challenges

Method	Description	Advantages	Challenges
In situ	Injecting CO ₂ into underground rock formations to react with minerals and form carbonates	Utilizes natural geological formations; scalable	Requires suitable geological sites; monitoring challenges
Ex situ	Capturing CO ₂ and reacting it with minerals or industrial by-products in a controlled setting	Flexible; can be tailored to specific needs	High energy costs; requires infrastructure
Bio-mineralization	Using microorganisms to enhance the reaction between CO ₂ and minerals to form carbonates	Potentially lower costs; can be applied to various environments	Slow reaction rates; require precise conditions
Enhanced Weathering	Spreading crushed silicate minerals on land to accelerate natural weathering processes that capture CO ₂	Large-scale potential; improves soil health	Large land use; dust and logistical issues

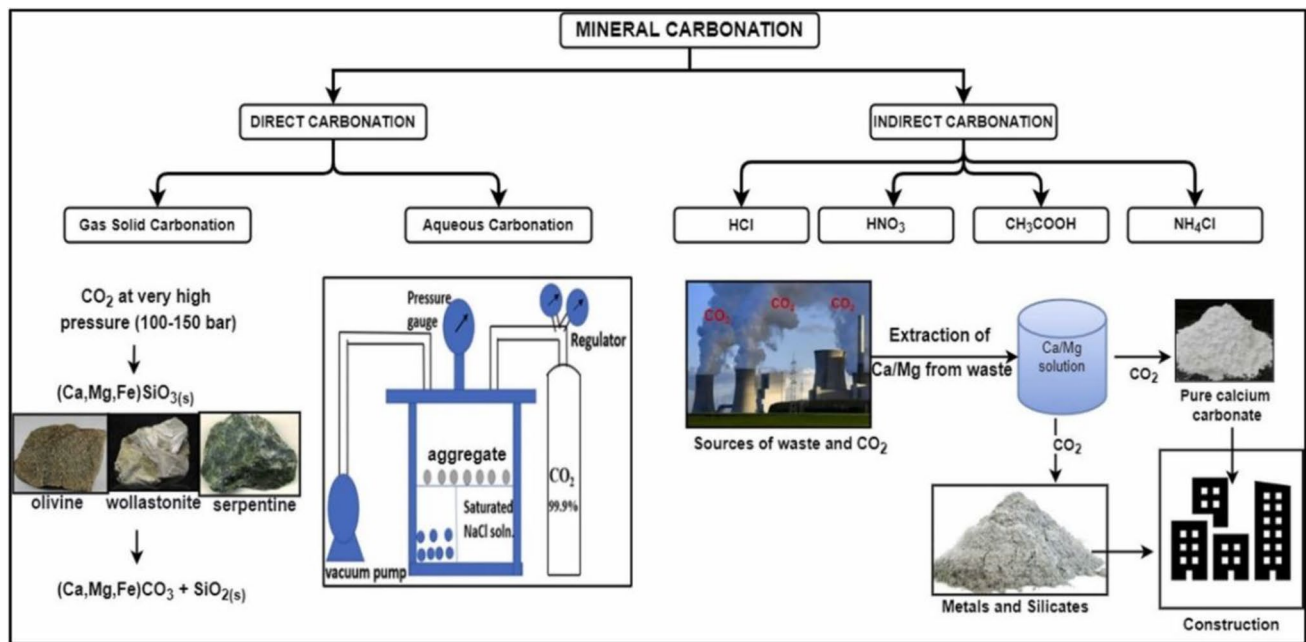


Fig. 7 Pathways of mineral carbonation [88]

4.4 Use of recycled aggregates

Incorporating construction and demolition (C&D) waste into low-carbon concrete is a sustainable practice that reduces the environmental impact of construction [89–91]. By recycling waste materials like crushed concrete, bricks, and asphalt, C&D waste can replace traditional aggregates, lowering the demand for virgin raw materials. This process decreases carbon emissions associated with concrete production, particularly by reducing cement consumption, which is a major contributor to CO₂ emissions. Additionally, using recycled materials diverts waste from landfills, promoting circular economy principles [92]. Overall, integrating C&D waste into low-carbon concrete helps create more eco-friendly construction solutions while maintaining material performance and structural integrity. The impact of using recycled materials on concrete strength and sustainability is twofold. In terms of strength, incorporating C&D waste can slightly reduce concrete's compressive strength compared to traditional mixes. However, this can be mitigated by optimizing mix design and incorporating additives like fly ash or silica fume. From a sustainability perspective, using recycled materials significantly reduces the environmental footprint [93, 94]. It lowers the demand for natural aggregates, reduces landfill waste, and cuts carbon emissions related to cement production. This makes concrete production more eco-friendly, supporting sustainable construction practices while balancing performance and durability.

4.5 Alternative binders

Literature highlights that OPC contributes approximately 8% of global CO₂ emissions, primarily from clinker production and fuel consumption during calcination. Strategies to mitigate these emissions focus on reducing clinker content, optimizing production processes, and utilizing alternative binders. Limestone Calcined Clay Cement (LC3) is a notable low-carbon cement that replaces a portion of clinker with calcined clay and limestone. Studies report a 30–40% reduction in CO₂ emissions compared to OPC [95]. The synergy between calcined clay and limestone enhances cement properties, including durability and resistance to chemical attacks, while requiring no significant changes in production infrastructure. Hydraulic Lime, a traditional binder, demonstrates potential as a low-carbon alternative due to its lower calcination temperature [96]. It emits less CO₂ during production and reabsorbs some carbon through carbonation over its lifecycle. However, its mechanical properties are generally inferior to OPC, limiting its use to specific applications. Magnesia Cement (MgO-based) has attracted interest for its ability to sequester

CO₂ during carbonation. Literature indicates that reactive magnesia binds with atmospheric CO₂, forming stable carbonates. Although promising, challenges include controlling carbonation rates and cost [97]. Research continues to optimize these materials, balancing performance and sustainability in cement production [98, 99].

5 Design strategies for low carbon concrete

Optimizing concrete mix design to reduce emissions focuses on balancing performance with sustainability. Supplementary cementitious materials (SCMs) like fly ash, slag, and silica fume help lower cement content, cutting carbon emissions. Using recycled aggregates and alternative binders such as geopolymers further reduces reliance on resource-intensive Portland cement. Improved water-cement ratios and chemical admixtures enhance strength and workability, minimizing waste. Advanced modelling and testing ensure optimal durability with fewer emissions. Figure 8 illustrates a four-stage framework for optimizing concrete mix design using data analysis and machine learning [100]. Stage 1 involves data acquisition and preprocessing, including tests and normalization. Stage 2 applies Bayesian optimization (BO) for predictive modelling. Stage 3 selects optimal mix proportions, and Stage 4 verifies algorithm accuracy through comparative analysis, SHAP interpretability, and microscopic testing.

Life Cycle Analysis (LCA) in low-carbon concrete evaluates its environmental impact throughout its life cycle, from raw material sourcing to disposal. Low-carbon concrete aims to reduce greenhouse gas emissions by using alternative materials like fly ash, slag, or recycled aggregates, minimizing energy-intensive cement production. Embodied carbon measurement quantifies the CO₂ emissions linked to producing and transporting concrete materials. By reducing the carbon footprint of cement (the main contributor), optimizing mix designs, and using carbon capture technologies, low-carbon concrete helps reduce the overall environmental impact. LCA ensures that these reductions are considered across the entire concrete life cycle.

Terán-Cuadrado et al. [48] highlighted the significance of material selection for low-carbon cement production. Their research identifies various materials that can substantially reduce carbon emissions during cement manufacturing, showcasing LCA as a strategic framework for optimizing material choices. Such insights are critical for engineers and architects aiming to enhance the sustainability of concrete structures, as they can now base their decisions on a thorough understanding of environmental impacts throughout the material's life cycle. Witte and Garg [101] further emphasized LCA's role in quantifying the global warming potential of different low-carbon concrete mixes. Their comparative analysis of existing LCA tools reveals considerable discrepancies in results, suggesting an urgent need for standardized

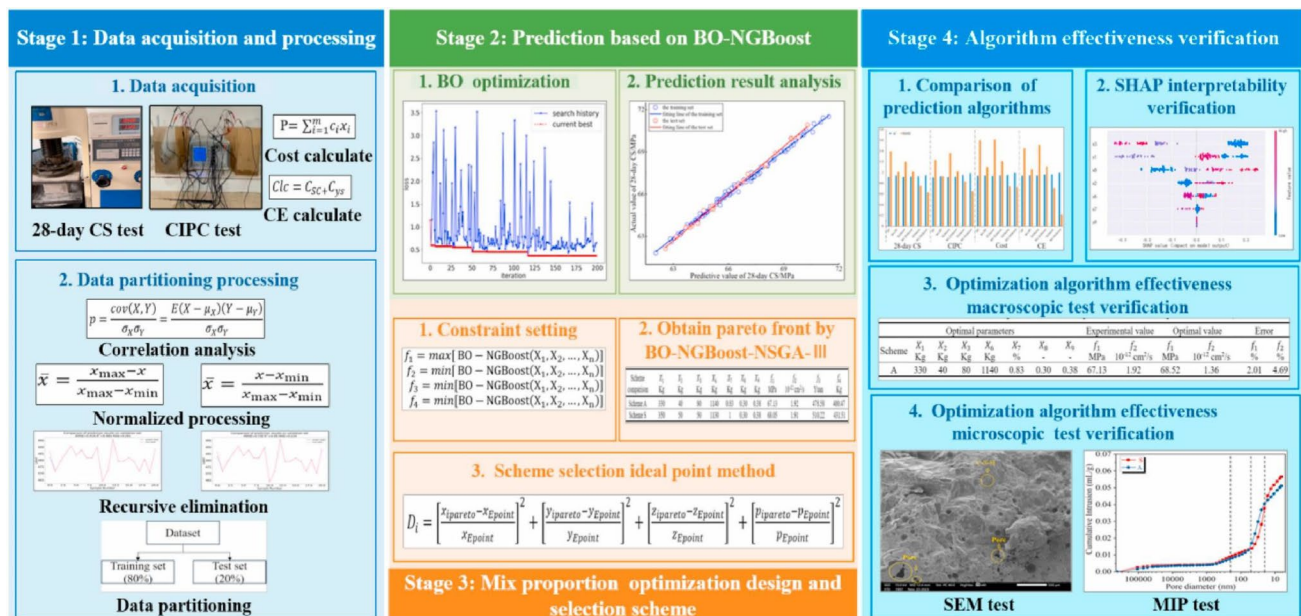


Fig. 8 Flow diagram of optimization of concrete mix proportion based on a hybrid intelligence method [100]

methodologies. The inconsistency in LCA outputs complicates decision-making for stakeholders, who require reliable metrics to evaluate the environmental benefits of various concrete options.

Innovative applications of materials in low-carbon concrete, as discussed by Wang et al. [102], illustrated the integration of advanced technologies with LCA practices. Their study demonstrates how experimental methods, including the use of microwave pre-curing with steel slag powder, can provide empirical data to enhance LCA models. This approach is vital for aligning theoretical assessments with practical applications, ultimately leading to improved environmental performance in concrete production. Additionally, Li et al. [103] explored the LCA of coal gangue composite cement, highlighting the potential of incorporating industrial by-products into low-carbon solutions. This example illustrates the versatility of LCA in evaluating unconventional materials, thereby promoting a circular economy approach in the construction industry.

6 Durability and performance considerations

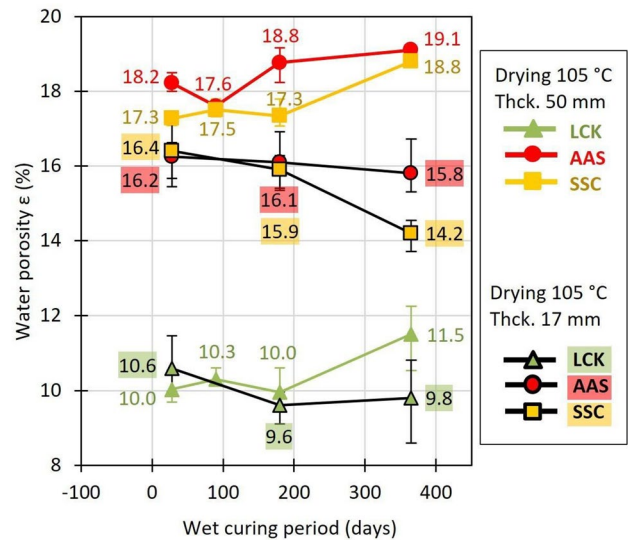
The long-term performance of low-carbon alternatives, such as low-carbon concrete or sustainable materials, focuses on their durability, functionality, and sustainability over extended periods. While these alternatives reduce greenhouse gas emissions during production, their effectiveness and longevity must be verified through rigorous testing and monitoring. Factors like material degradation, structural integrity, and maintenance needs are crucial in assessing performance. Ensuring that low-carbon alternatives maintain their quality and functionality over time is essential for their widespread adoption. Long-term performance evaluations help confirm that these materials not only offer environmental benefits but also meet the practical demands of construction and usage.

In low-carbon concrete, durability issues like corrosion, cracking, and shrinkage can impact its performance. Corrosion can arise if alternative materials or reduced cement content affect the concrete's resistance to moisture, compromising the protection of reinforcement bars. Cracking may occur due to altered mix designs or differences in curing practices, potentially reducing structural integrity. Shrinkage, influenced by variations in material properties or mix composition, can lead to fissures and surface defects. Addressing these concerns involves optimizing mix formulations, using advanced admixtures, and implementing robust construction practices. Ensuring durability is essential for maintaining the long-term effectiveness and safety of low-carbon concrete structures.

Durability and performance considerations are critical in evaluating low-carbon concrete alternatives, especially in light of increasing environmental concerns and the need for sustainable construction materials. The research surrounding low-carbon concrete, particularly through the integration of supplementary cementitious materials (SCMs) and innovative formulations, reveals both promising advancements and challenges. Engineered geopolymer composites have emerged as a viable option for creating ultra-high-performance concrete (UHPC) with lower carbon emissions. According to Elmesalami and Celik [104], these materials exhibit superior mechanical properties and durability compared to traditional concrete. However, their long-term performance, particularly under varied environmental conditions, necessitates further investigation. The durability of these composites can be affected by factors such as microstructural integrity and the types of raw materials used, making it essential to optimize formulations to ensure lasting performance.

The focus on low-cement concrete highlights the importance of time-dependent properties and overall durability. Robalo et al. [105] discussed how reducing cement content can effectively lower carbon footprints, but this reduction may also introduce vulnerabilities to mechanical performance over time. Their findings suggest that careful selection of SCMs is crucial for enhancing durability, as these materials significantly influence resistance to environmental factors such as freeze–thaw cycles and chemical attacks. Furthermore, Kumar et al. [66] investigated the impact of various SCMs on the physico-mechanical and durability characteristics of low-carbon cement composites. Their comprehensive study indicates that while SCMs like fly ash and slag can enhance performance, their effectiveness is heavily dependent on the proportions used and the specific environmental conditions encountered. This underscores the necessity of developing tailored mixes that account for both initial performance and long-term durability. Another critical aspect is the reinforcement of low-carbon UHPC with multiscale steel fibres, as explored by Cui et al. [106]. Their research demonstrates that fibre reinforcement can significantly enhance durability by mitigating crack propagation, addressing one of the common failures in low-carbon mixes. This suggests a promising pathway for improving the resilience of these materials, potentially leading to more robust and durable concrete solutions. Additionally, research by Zhang et al. [107] into low-carbon geopolymer mortars emphasized the varying responses of these materials to cryogenic attacks based on water content and freeze–thaw conditions. This highlights the importance of understanding the environmental interactions that can affect durability, particularly in extreme climates, which is essential for ensuring the longevity of low-carbon concrete.

Fig. 9 Water porosity, ϵ , for LCK (in green), AAS (in red), and SSC (in yellow) concretes after 28 days, 90 days, 180 days, and 365 days of wet curing, with a drying step of either 105 °C and a sample thickness of 50 mm (colored lines) or 50 °C and 17 mm (black lines) [108]

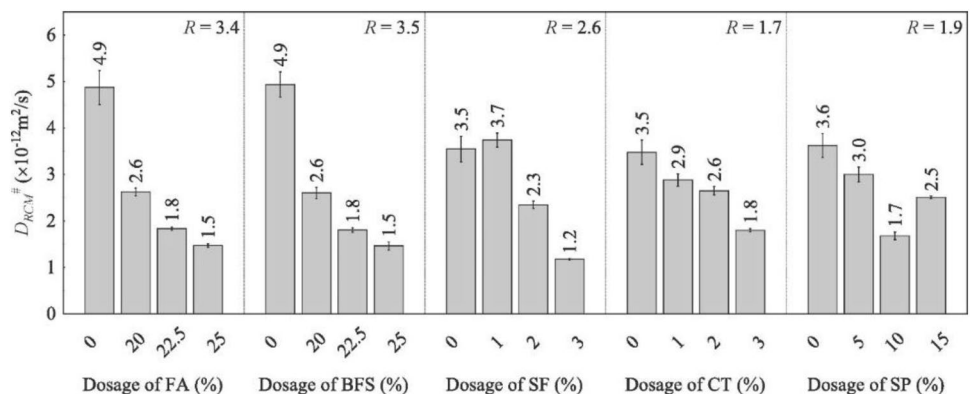


Results of water porosity with a drying step at 105 °C or 50 °C are presented in Fig. 9 for different periods of curing [108]. The figure shows the relationship between wet curing duration and water porosity in three types of materials low clinker (LCK) concrete, alkali-activated (AAS) concrete, and super-sulphated cement (SSC) concrete at two thicknesses (50 mm and 17 mm), after drying at 105 °C. For the thicker samples (50 mm), all materials show relatively high porosity, which generally increases or stabilizes over time. AAS has the highest porosity, peaking at 18.8% after 200 days and further rising to 19.1% by 400 days. SSC follows a similar but slightly lower trend, peaking at 17.6% and ending at 18.8%. LCK shows a minor increase in porosity over time, reaching 15.8% at 400 days, the lowest among the three materials at this thickness. For the thinner samples (17 mm), porosity values are significantly lower across all materials. LCK starts with a lower porosity and decreases to 9.6% by 200 days, stabilizing near 9.8% at 400 days, showing effective densification with curing. AAS and SSC display an initial decrease, reaching 10.3% and 10.0%, respectively, by 200 days. Overall, LCK shows the lowest porosity at both thicknesses, suggesting a more impermeable structure. Thinner samples for all materials exhibit lower porosity than thicker ones, indicating that reduced thickness enhances densification and reduces porosity, possibly due to more efficient moisture removal and curing in thinner sections.

Low-carbon concrete differs from conventional concrete primarily in its reduced environmental impact. It incorporates alternative materials like fly ash or slag, which lower the carbon footprint compared to traditional cement. While conventional concrete relies heavily on Portland cement, contributing to high CO₂ emissions, low-carbon concrete aims to minimize these emissions through innovative mix designs and material substitutions. Durability and performance can be comparable if properly formulated, though low-carbon options may require adjustments in curing and handling. Overall, low-carbon concrete offers a more sustainable choice with the potential for similar or enhanced structural properties while significantly reducing environmental impact.

Figure 10 illustrates the impact of various factors on the chloride migration coefficient of Fair-faced Concrete (FFC) [109]. The dosage of Blast Furnace Slag (BFS) is the most significant factor, followed by Fly Ash (FA), Silica Fume (SF),

Fig. 10 Chloride migration coefficient for FFCs with different mineral admixtures at different dosages [109]



Superplasticizer (SP), and Curing Time (CT). For optimal resistance to chloride ion penetration, a mixture of 25% FA, 25% BFS, 3% SF, 3% CT, and 10% SP is recommended. Incorporating FA and BFS can reduce the chloride migration coefficient by up to 70% due to their pozzolanic reactivity, which enhances compactness and resistance to chloride ingress. While 1% SF has minimal effect, increasing SF to 2% or 3% significantly reduces the coefficient by up to 67.6%, improving compactness due to its spherical shape and reactivity.

7 Environmental and economic benefits of low carbon concrete

Low carbon concrete, an innovative alternative to traditional concrete, offers significant CO₂ reduction potential. This environmentally friendly concrete reduces carbon emissions through several methods, such as incorporating industrial byproducts like fly ash or slag, using alternative binders like geopolymer, or optimizing the mix design to lower the clinker content. Additionally, advancements in carbon capture and storage technologies can further enhance its sustainability. By lowering the carbon footprint associated with cement production—one of the largest industrial sources of CO₂—low carbon concrete helps mitigate climate change and supports greener construction practices, making it a crucial component in sustainable building strategies. The cost–benefit analysis of low-carbon concrete solutions reveals a complex interplay between initial investments, long-term savings, and environmental impacts. Liu and Li [110] emphasized the economic benefits of managing construction waste through carbon trading policies, indicating potential financial gains from adopting low-carbon technologies. Additionally, Yosri et al. [111] explored the valorization of engineered biochar in ultra-high-performance fibre-reinforced concrete, demonstrating that integrating sustainable materials can yield significant reductions in carbon emissions while enhancing performance. Chen et al. [112] further supported this, showing that biochar-augmented concrete can achieve carbon negativity, thereby offsetting emissions. Meanwhile, Sadok et al. [113] assessed the performance and economic viability of low-carbon binders derived from calcined sediments, highlighting innovative approaches to material sourcing that can reduce costs while promoting sustainability. Collectively, these studies underscore the financial and environmental advantages of low-carbon concrete, advocating for broader adoption within the construction industry.

Table 3 highlights a growing trend toward low-carbon concrete solutions, reflecting increasing initial costs alongside significant long-term savings and environmental benefits. While traditional concrete remains less expensive upfront, low-carbon alternatives like engineered geopolymer and fibre-reinforced concrete offer enhanced durability and reduced lifecycle emissions, ultimately leading to greater sustainability. The economic benefits, including potential job creation and competitive advantages in green building markets, further bolster the appeal of low-carbon technologies. Overall, the trend indicates a shift towards integrating environmental considerations into construction practices, aligning with global sustainability goals and responding to rising regulatory pressures. The three plots (Fig. 11) compare various concrete types across costs, long-term savings, and environmental impacts. The first plot highlights initial, material, and labour costs, showing variations by type. The second plot emphasizes potential long-term savings, with high-performance options offering greater benefits. The third plot demonstrates environmental impacts, with lower emissions for sustainable alternatives.

8 Challenges and barriers to adoption

Large-scale production of low-carbon concrete faces several technical challenges. Firstly, the need for alternative materials, such as supplementary cementitious materials or recycled aggregates, can affect the mix's performance and consistency [124–126]. Ensuring uniformity and quality control at scale is difficult. Additionally, integrating new technologies or methods, like carbon capture and storage, requires significant infrastructure and investment [127–129]. The curing process may also need adjustments to accommodate new compositions. Addressing these challenges requires advancements in material science, process optimization, and supply chain management, alongside rigorous testing to ensure that low-carbon concrete meets performance standards while reducing environmental impact.

Regulatory and standardization issues are crucial in the widespread adoption of low-carbon concrete. Different regions may have varying building codes and standards, which can complicate the approval and implementation of new concrete formulations. Lack of uniform standards can lead to inconsistencies in quality and performance. Additionally, existing regulations may not account for the unique properties of low-carbon concrete, necessitating updates or new guidelines. Ensuring that low-carbon concrete meets safety, durability, and performance requirements while aligning with diverse

Table 3 Cost–benefit analysis of low-carbon concrete solutions

Aspect	Initial costs	Material costs	Labor costs	Long-term savings	Environmental benefits	Economic benefits	References
Traditional Concrete	\$130–\$200 per cubic meter	\$100–\$150 per cubic meter	\$25–\$40 per cubic meter	Minimal long-term savings; maintenance costs apply	High carbon footprint; approximately 0.9 tons CO ₂ per cubic meter	Lower initial costs but higher maintenance expenses	[114, 115]
Low-Carbon Concrete	\$140–\$230 per cubic meter	\$90–\$140 per cubic meter	\$20–\$30 per cubic meter	\$15–\$30 savings per cubic meter in energy costs	Reduced carbon emissions by 30–50%; about 0.5 tons CO ₂ per cubic meter	Competitive advantage in green building certifications	[116, 117]
High-Performance Low-Carbon Concrete	\$180–\$300 per cubic meter	\$120–\$180 per cubic meter	\$35–\$55 per cubic meter	\$25–\$40 savings in reduced repairs and maintenance	Lower lifecycle emissions and improved durability	Increased lifespan of structures leading to cost avoidance	[118, 119]
Fibre-Reinforced Low-Carbon Concrete	\$170–\$270 per cubic meter	\$110–\$170 per cubic meter	\$30–\$50 per cubic meter	Reduced maintenance costs due to enhanced durability	Better resistance to cracking reduces waste and emissions	Job creation in specialized fibre manufacturing and application	[120, 121]
Recycled Aggregate Concrete	\$150–\$240 per cubic meter	\$70–\$110 per cubic meter	\$20–\$30 per cubic meter	\$10–\$20 savings from using recycled materials	Diverts waste from landfills and lowers resource extraction impact	Lower cost for material procurement and local sourcing benefits	[122, 123]

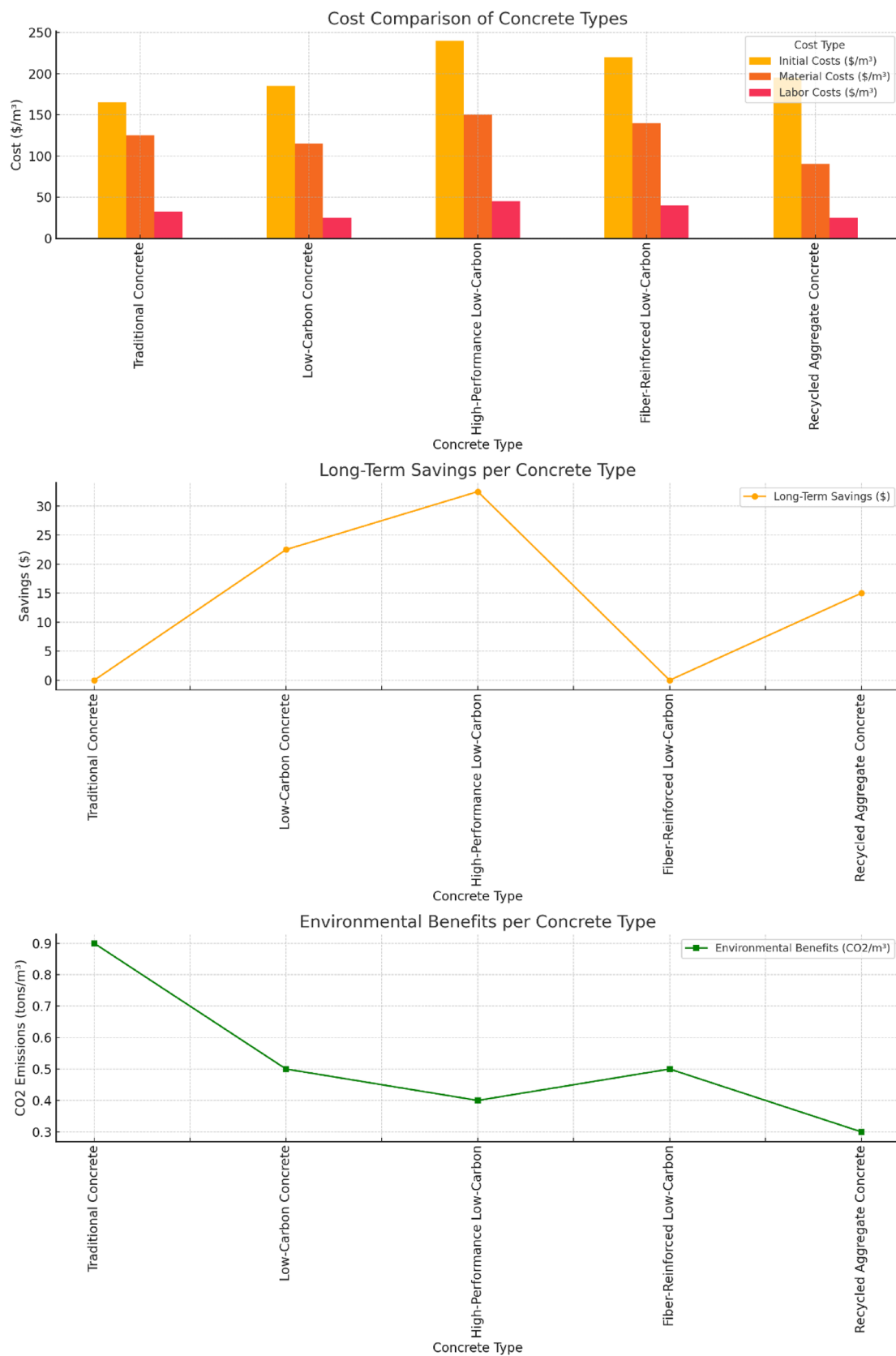


Fig. 11 Comparative analysis of concrete types: costs, savings, and environmental impact

regulations is essential [130–132]. Collaborative efforts among policymakers, industry experts, and standard organizations are vital to address these challenges effectively.

Market acceptance and adoption rates of low-carbon concrete are gaining momentum as the construction industry seeks sustainable alternatives. This eco-friendly material, which reduces carbon emissions compared to traditional concrete, is increasingly favoured due to growing environmental regulations and heightened awareness of climate change. Early adopters, such as progressive construction firms and green building projects, are driving its uptake. However, widespread adoption faces hurdles like higher costs and the need for new standards and supply chains. As technology advances and economies of scale improve, low-carbon concrete is poised to become a mainstream choice, aligning with global sustainability goals.

Supply chain and resource availability are critical factors influencing the production and adoption of low-carbon concrete. Ensuring a steady supply of materials like SCMs, which reduce carbon footprints, is essential for consistent product quality. Challenges include sourcing and processing SCMs, managing transportation logistics, and overcoming regional shortages. Innovations in material sourcing and advancements in recycling technologies can help mitigate these issues. Developing robust supply chains and improving resource management are crucial for scaling low-carbon concrete production and making it more accessible to a broader market, thereby supporting global sustainability initiatives.

9 Future trends and research directions

Advances in material science and technology have revolutionized low-carbon concrete, aiming to reduce its environmental impact. Key innovations include the development of alternative cementitious materials like fly ash, slag, and silica fume to replace traditional Portland cement, which is energy-intensive and CO₂-heavy. Carbon capture techniques are also integrated, trapping CO₂ during production. Novel additives and admixtures enhance strength and durability, reducing the need for excessive material use [133–135]. Additionally, 3D printing and self-healing concrete are improving design precision and longevity, further minimizing waste and emissions [136, 137]. These innovations are driving the construction industry toward more sustainable, carbon-efficient solutions.

Emerging trends in carbon capture and low-carbon materials are shaping the future of sustainable industries. Innovations in carbon capture include direct air capture (DAC) technologies, which remove CO₂ directly from the atmosphere, and bioenergy with carbon capture and storage (BECCS), using biomass to capture carbon [138, 139]. Additionally, low-carbon materials are gaining prominence, such as carbon-neutral concrete made with alternative binders and recycled materials, and bioplastics derived from renewable resources [140, 141]. Advances in using captured CO₂ to produce carbon-negative products, like synthetic fuels and construction materials, are also key. These trends are critical for reducing industrial emissions and supporting climate goals.

Policy and regulatory frameworks are essential for encouraging the adoption of sustainable technologies and practices. Governments can implement carbon pricing mechanisms, such as carbon taxes and cap-and-trade systems, to create economic incentives for reducing emissions [142–144]. Subsidies, grants, and tax credits for renewable energy projects and carbon capture technologies also drive investment in low-carbon solutions. Setting stringent emissions standards and establishing clear guidelines for sustainable practices across industries encourage compliance. Furthermore, creating green certification programs and providing financial support for research and development foster innovation [145, 146]. (International cooperation and aligning policies with global climate targets are crucial for accelerating the transition to a low-carbon economy.

Digital technologies such as artificial intelligence (AI) and the Internet of Things (IoT) are essential for advancing low-carbon concrete design. AI enhances material optimization and predictive modelling, enabling improved concrete mix designs that reduce carbon emissions. It identifies the ideal balance of materials to achieve both performance and sustainability. Meanwhile, IoT sensors offer real-time monitoring of critical parameters like curing conditions and energy consumption during production. This ensures efficient processes and optimal concrete performance. When integrated, AI and IoT create powerful, adaptive systems for advanced data analysis and decision-making. AI processes the data collected by IoT sensors, enabling precise adjustments to improve concrete quality and minimize its environmental impact. This synergy not only optimizes production but also enhances sustainability throughout the concrete's lifecycle. These technologies provide builders and manufacturers with tools to reduce carbon footprints while maintaining cost efficiency and performance. By enabling precise control and smart systems, AI and IoT pave the way for greener, more eco-friendly construction solutions. Their combined capabilities represent a transformative step toward achieving sustainability goals in the construction industry, making environmentally responsible practices both feasible and practical.

Machine learning (ML) is revolutionizing the prediction of low-carbon concrete performance by analyzing vast datasets to identify patterns and optimize mix designs. Traditional methods rely on time-consuming laboratory tests, while ML models, such as artificial neural networks (ANNs) and support vector machines (SVMs), can accurately predict compressive strength, durability, and carbon footprint [147–150]. These models utilize input parameters like water-to-binder ratio, supplementary cementitious materials, and curing conditions. By leveraging big data, ML enhances sustainability efforts, reduces material waste, and accelerates innovation in green construction. Ultimately, ML-driven insights help engineers develop more efficient and eco-friendly concrete solutions for a sustainable future.

Graphene quantum dots (GQDs) are nanoscale materials derived from graphene, offering remarkable properties such as high electrical conductivity, thermal stability, and photoluminescence. In cement composites and concrete, GQDs enhance mechanical and functional performance [145, 151, 152]. They improve compressive and tensile strength by refining microstructures and promoting effective hydration [153, 154]. Their inclusion also boosts crack resistance, durability, and toughness. Moreover, GQDs provide self-sensing capabilities, enabling real-time structural health monitoring. Their nanoscale dispersion enhances concrete's thermal and electrical conductivity, contributing to energy-efficient construction. Despite these benefits, challenges like cost and scalability need addressing to fully harness GQDs in sustainable, high-performance concrete for modern infrastructure.

10 Concluding remarks

Low-carbon concrete represents a transformative innovation for reducing the construction sector's environmental footprint and aligning with global sustainability objectives. This review underscores the pressing need for alternatives to traditional concrete, highlighting its substantial carbon footprint. Viable solutions include supplementary cementitious materials (SCMs), alkali-activated and geopolymer concrete, carbon capture and utilization (CCU), and recycled aggregates, each offering the potential to lower emissions without sacrificing performance. However, adoption challenges such as technical barriers, regulatory limitations, and resource availability must be addressed to ensure widespread implementation.

Future research should focus on optimizing material performance, lifecycle analyses, and scalability of innovative concrete technologies. Collaboration between academia and industry can drive breakthroughs in integrating advanced tools like artificial intelligence (AI) and the Internet of Things (IoT) for efficient mix designs and process optimization. Furthermore, governments must establish robust policy frameworks and incentives to promote low-carbon solutions, encouraging investment and adoption at a global scale. Looking ahead, low-carbon concrete holds immense potential to reshape the construction industry, offering a pathway to significantly reduce emissions, enhance environmental sustainability, and contribute to achieving international climate goals. Coordinated efforts across sectors will be pivotal in accelerating progress and fostering a sustainable future.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate This review was conducted in accordance with ethical standards for academic research and publication. The authors confirm that no human participants or animals were involved in the creation of this review paper, and therefore, ethical approval was not required. All sources and references have been properly cited to acknowledge the contributions of other researchers. The authors have adhered to best practices for transparency, integrity, and objectivity in the preparation and presentation of this review.

Competing interests The authors declare no competing interests.

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